NAME

pmod_stream – Pmod pin-stream player

SUMMARY

Catalog IP **user.org:user:pmod_stream:1.0 .** AXI4-Lite slave, 64 KB window, 8-bit bidirectional Pmod port. The player consumes a run-length stream from block RAM and drives the pins on **play_clk .** Software fills the stream. It does not bit-bang the pins.

Item	Value
Play timebase	play_clk, 200 MHz in the Cmod S7 reference system
AXI clock	s_axi_aclk, independent of play_clk
Port width	8 bits, per-pin output enable
TX memory	8192 words (32 KB) at the default parameter
Capture	8192 samples, one byte per played word
Word format	{OE[7:0], DOUT[7:0], duration[15:0]}
Duration	1 to 65535 play_clk cycles (5 ns to 327.675 us at 200 MHz)
Sample point	last cycle of each word, after a 2-flop synchronizer
Idle	IDLE_OE and IDLE_OUT while not playing. Reset forces Hi-Z
Interrupt	level high, done or fault, while CTRL.irq_en is set

DESCRIPTION

A Pmod header is an 8-bit digital connector. Protocols such as SPI, I2C, UART, and timed GPIO are produced by writing a list of pin states and how long each state lasts. **pmod_stream** walks that list on **play_clk** and records the pin levels at the end of every entry.

While the player is idle, the pins follow **IDLE_OE** and **IDLE_OUT .** A 1 in **IDLE_OE** drives the matching **IDLE_OUT** bit. A 0 releases the pin. Both registers reset to zero, so the port comes out of reset Hi-Z. The end of a play, including a fault and an abort, restores those idle levels.

Open-drain protocols are a property of the stream, not a pin mode in the IP. A zero is written as output-enable 1 and data 0. A one is written as output-enable 0. The player never has to drive a high on that pin. An external or on-board pull-up supplies the high level.

PORTS

Port	Direction	Clock	Description
s_axi_*	**slave**	**s_axi_aclk**	AXI4-Lite, 16-bit address, 32-bit data
s_axi_aclk	**in**	**-**	Register and memory port A clock
s_axi_aresetn	**in**	**s_axi_aclk**	Active-low AXI reset
play_clk	**in**	**-**	Player timebase
play_aresetn	**in**	**async, synced to play_clk**	Active-low player reset
pmod_io[7:0]	**inout**	**play_clk**	Pmod data pins
irq	**out**	**s_axi_aclk**	Level-high interrupt
status_led	**out**	**s_axi_aclk**	Busy, or a heartbeat while idle

s_axi is associated with **s_axi_aclk** and **s_axi_aresetn . play_clk** is associated with **play_aresetn .** The two clocks are asynchronous. Start, abort, done, idle levels, busy, and the pin sample cross between them. **irq** is level high. It stays asserted while **CTRL.irq_en** is set and either **STATUS.done** or **STATUS.fault** is set.

PARAMETERS

Name	Default	Legal	Meaning
TX_WORDS	**8192**	**up to 8192**	**Number of stream words in the TX RAM**
RX_SAMPLES	**8192**	**multiple of 4, up to 8192**	**Number of captured pin bytes**

The address map below is fixed for the default sizes. **TX_WORDS** above 8192 does not fit the 13-bit word index. **RX_SAMPLES** is packed four bytes per AXI word, so it must be a multiple of four. The packaged default is the size used by the PModPirate reference system.

STREAM WORD

Each TX location is one 32-bit little-endian word. A write must have all four byte strobes set. A partial strobe is ignored.

Bits	Field	Meaning
31:24	**OE**	**1 = drive pmod_io[i], 0 = Hi-Z**
23:16	**DOUT**	**Level driven where OE is 1**
15:0	**DURATION**	**play_clk cycles this word is held, minimum 1**

The word is applied on a rising edge of **play_clk** and remains on the pins for **DURATION** cycles. On the edge where that count expires, the player stores the synchronized pin sample and either applies the next word or returns to idle. The last word is not held for an extra cycle.

A duration of 0 is a fault. The play stops and the pins return to idle. Holds longer than 65535 cycles are several words with the same OE and DOUT. At 200 MHz the longest single word is 327.675 us.

CAPTURE

On the last cycle of each word the player writes one byte into the capture RAM. Bit *i* of that byte is **pmod_io[i]** after two **play_clk** flops. The byte for word *n* is stored at sample index *n* . **CAP_COUNT** reports how many samples the play stored.

Because the sample passes through two flops, a level that changes on the same edge that starts a very short word can still show the previous level. A word whose duration is at least 4 play clocks shows a level that became valid at the start of that word. Place a word boundary on the instant that matters, and give that word enough cycles for the synchronizer. Mid-bit of a UART character, the clock-high side of SPI, and the SCL-high window of an I2C ACK are the usual places. Samples from other words are ignored by the consumer.

Four samples are packed into one AXI word, little-endian. Sample *i* is byte *(i mod 4)* of the word at the RX base plus *4 * (i / 4)* . Byte 0 is bits 7:0.

REGISTER MAP

Offsets are bytes from the base of the slave. The slave window is 64 KB. In the Cmod S7 reference system that base is **0x00010000 ,** directly above the 64 KB MicroBlaze local memory.

Offset	Name	Access	Description
0x0000	**CTRL**	**R/W**	bit 0 start, bit 1 abort, bit 2 irq_en
0x0004	**STATUS**	**R/W1C**	bit 0 busy, bit 1 done, bit 2 fault
0x0008	**LENGTH**	**R/W**	Number of TX words to play, bits 15:0
0x000C	**CAP_COUNT**	**R**	Samples stored by the last play, bits 15:0
0x0010	**IDLE_OUT**	**R/W**	Idle drive data, bits 7:0
0x0014	**IDLE_OE**	**R/W**	Idle output enable, bits 7:0. 1 = drive
0x0018	**PIN_IN**	**R**	Synchronized pin sample, bits 7:0
0x1000	**TX**	**R/W**	8192 words, through 0x8FFC
0x9000	**RX**	**R**	2048 packed words, through 0xAFFC

Reserved bits read as 0. Writes to reserved bits are ignored. AXI responses are OKAY. **CTRL** reads back **irq_en** in bit 2. Start and abort are pulses, so those bits read as 0.

STATUS.done and **STATUS.fault** are sticky. Write 1 to the bit to clear it. Writing 0 leaves it. **STATUS.busy** is read-only.

PIN_IN is the capture synchronizer crossed back to **s_axi_aclk .** It is the register for a GPIO read that does not need a timed sample. It is not a substitute for the capture RAM during a play.

TX reads and writes use the full 32-bit word. RX is read-only. A TX write with **WSTRB** other than **0xF** does not change memory.

PROGRAMMING A PLAY

1. Write the words at **TX ,** starting at offset 0x1000.
2. Write **LENGTH** with the word count. Do this before start. The value is copied across the clock domain with the start pulse.

3. Clear the previous result by writing **0x6** to **STATUS .**
4. Write **CTRL** bit 0. Bit 2 may be set in the same write to enable the interrupt.
5. Poll **STATUS** until **done** is set and **busy** is clear, or wait for **irq .**
6. If **fault** is clear, read **CAP_COUNT** and the packed samples at **RX .**

A second start while **busy** is set, or while a start is still crossing into **play_clk ,** sets **fault** and does not start another play. Wait for **done** before starting again.

Abort by writing **CTRL** bit 1. The player returns to idle, sets **done ,** and does not set **fault** for the abort itself.

FAULTS

STATUS.fault is set, the pins return to idle, and **done** is set, when any of the following happens.

- **LENGTH** is 0 or greater than **TX_WORDS .** No word is driven.
- A played word has duration 0. Words before it were driven.
- The next word is not available on the cycle a word expires. That is an underrun. With the default memory and a prefetch queue of four words it is not expected for a legal duration of at least 1.
- **CTRL.start** is written while a play is already active or a start is still pending.

CAP_COUNT then reports how many samples were stored before the stop. An abort is not a fault.

CLOCK, RESET, AND STATUS LED

play_aresetn is synchronized into **play_clk** before it resets the player. **s_axi_aresetn** resets the AXI registers directly. Either reset clears output enable, so the pins are Hi-Z, and clears **irq_en , done ,** and **fault .**

Idle levels use a multi-flop array synchronizer. A write to **IDLE_OE** or **IDLE_OUT** is visible on the pins a few **play_clk** cycles later, and only while the player is idle. During a play the stream overrides them.

status_led is high while **STATUS.busy** is set. While idle it follows bit 25 of a counter on **s_axi_aclk .** At 100 MHz that bit is high for about 0.34 s and low for about 0.34 s.

REFERENCE SYSTEM

On the Cmod S7-25 design the slave is at **0x00010000** and the UART is a separate peripheral. **s_axi_aclk** is 100 MHz. **play_clk** is 200 MHz. Both come from an MMCM whose input is the 12 MHz board clock. The interrupt is connected to the MicroBlaze interrupt controller. The firmware polls **STATUS** and leaves **irq_en** clear. The eight Pmod pins are LVCMOS33 with pull-ups. The status LED is a board LED.

Those choices belong to the board design. The IP itself only requires a stable **play_clk ,** an AXI clock, and the two active-low resets.

UTILIZATION

Out-of-context synthesis of the default configuration, Vivado 2025.2, part xc7s25csga225-1:

Resource	Used	Device
Slice LUTs	**361**	**14600**
Slice registers	**542**	**29200**
RAMB36	**16**	**45**
RAMB18	**4**	**90**
DSP	**0**	**80**

The block RAM holds the 32 KB TX stream and the 8 KB capture buffer. The count is what inference produced for the true dual-port TX RAM and the four byte-wide capture banks. It is a synthesis result, not a placed-and-routed total for the whole design.

REVISION

Version	Date	Notes
1.0	**2026-11-07**	First packaged player. Default 8192 words and 8192 samples.